

Technical Report

Credit Default Risk Analysis

Date: May 26, 2026

1. Executive Summary

This report details the development of a predictive model for credit card default risk using historical data from 30,000 clients. Our final ensemble model identifies high-risk borrowers with significant precision, allowing for proactive risk mitigation.

2. Methodology & Approach

Given approach focused on transforming raw billing behavior into actionable risk signals.

Data Quality & Preprocessing

- **Categorical Encoding:** Addressed undocumented values in `EDUCATION` (0, 5, 6) and `MARRIAGE` (0) by merging them into an "Other" category to maintain data integrity without losing sample size.
- **Anomaly Handling:** Identified negative `BILL_AMT` values (indicating overpayments). These were kept as they reflect valid account states but were clipped during utilization calculations to prevent negative ratios.
- **Class Imbalance:** With defaults at ~22%, I utilized `class_weight='balanced'` and `scale_pos_weight` to ensure the model prioritized the minority class (actual defaulters).

Feature Engineering

Beyond demographic data, I engineered three "behavioral" features:

1. **AVG_UTIL_RATE:** Measures how much of the credit line is consistently used. High utilization is a classic precursor to default.
2. **AVG_PAY_RATIO:** Captures repayment consistency. A dropping ratio indicates liquidity stress.
3. **TOTAL_DELAY_MONTHS:** A cumulative delinquency counter that quickly flags chronic late payers.

3. Model Selection & Tradeoffs

Here, we evaluated two distinct architectures:

- **Logistic Regression (Baseline):** Chosen for its transparency and speed. However, it struggled with the non-linear relationships between payment history and risk.
- **Tree-Based Ensemble (XGBoost/Random Forest):** These models captured complex interactions (e.g., the combined risk of high age and high utilization).

Tradeoffs:

- While Logistic Regression is easier to explain to regulators, the **XGBoost** model provided a ~15% improvement in AUC-ROC, which translates to significantly fewer missed defaults (lower False Negatives).

4. Key Findings

- **Signal Strength:** The most recent payment status (`PAY_0`) is the single most important predictor. A delay in the current month is the highest probability indicator of default in the next.
- **Risk Thresholds:** Clients with more than 2 months of cumulative delay (`TOTAL_DELAY_MONTHS`) are 3x more likely to default than those with consistent on-time payments.

5. Strategic Recommendations

1. **Dynamic Limit Adjustments:** Implement automated triggers to reduce `LIMIT_BAL` for clients whose `AVG_UTIL_RATE` exceeds 80% while showing a `PAY_0` status of 1+.
2. **Early Intervention:** Use the `TOTAL_DELAY_MONTHS` signal to launch "soft" collection outreach (reminders) before the client reaches a 60-day delinquency state.

6. Future Improvements

Given more time, Following updates would be done:

- **Hyperparameter Optimization:** Use *BayesianSearch* or *RandomizedSearch* to fine-tune the XGBoost model for even higher Recall.
- **SHAP Global Interpretability:** Develop a dashboard showing *why* specific clients are flagged, ensuring the credit team can provide specific feedback to clients.
- **Combining Ensemble models :** It boosts predictive accuracy, reducing overfitting, and increasing overall robustness.
- **Post Processing Optimization :** Improves the reliability, and business effectiveness of model predictions without retraining the model.